University of Dhaka
Department of Nuclear Engineering
3rd Year 2nd Semester B.Sc. (Engg.) 2023
Course Code: NE-3204
MATLAB+LabVIEW
Assignment, Submission Date: September 17, 2023

Time: N/a Full Marks: 30

Answer all questions

1. Solve the following 'idealized' diffusion domain on the dummy variable ξ,

$$\frac{\partial \xi}{\partial t} + \frac{\partial^2 \xi}{\partial x^2}(x, y) + \frac{\partial^2 \xi}{\partial y^2}(x, y) = 0$$

in a 2D Cartesian coordinate system using FDM for

(a) steady-state and **(10)**

(b) transient. **(20)**

If it isn't obvious enough, drop the temporal term for steady-state. Both the steady-state and the transient cases should produce the same solution (after you've run the transient simulation for 'some time', tha t is).

See Fig. (1) for domain definition.

Use forward differencing in time and central differencing in space,

$$\frac{\partial \xi}{\partial t} = \frac{\Xi_{i,j}^{n+1} - \Xi_{i,j}^{n}}{\Delta t}$$
$$\frac{\partial^2 \xi}{\partial x^2} = \frac{\Xi_{i+1,j}^{n} - 2\Xi_{i,j}^{n} + \Xi_{i-1,j}^{n}}{(\Delta x)^2}$$
$$\frac{\partial^2 \xi}{\partial y^2} = \frac{\Xi_{i,j+1}^{n} - 2\Xi_{i,j}^{n} + \Xi_{i,j-1}^{n}}{(\Delta y)^2}$$

and solve for $\Xi_{i,j}^{n+1}$.

I am setting a Dirichlet BC. Heed the indexing paradigm for discretization; see Fig. (2).

Write iterative solvers based on: (i) Jacobi, (ii) Gauss–Seidel, and (iii) SOR. Compare your solver solutions with the built-in "\" (backslash) operator solution.

Set your convergence criterion based on absolute or relative error (or Euclidean/ℓ_2 norm or anything else).

You are advised to present your results in contour/temperature plots.

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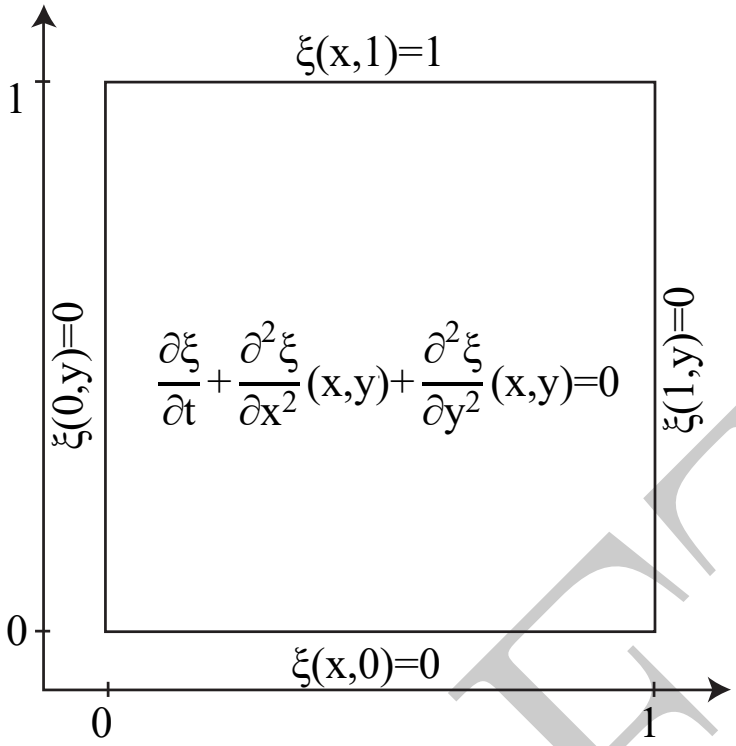


Figure 1: Domain with Dirichlet BC.

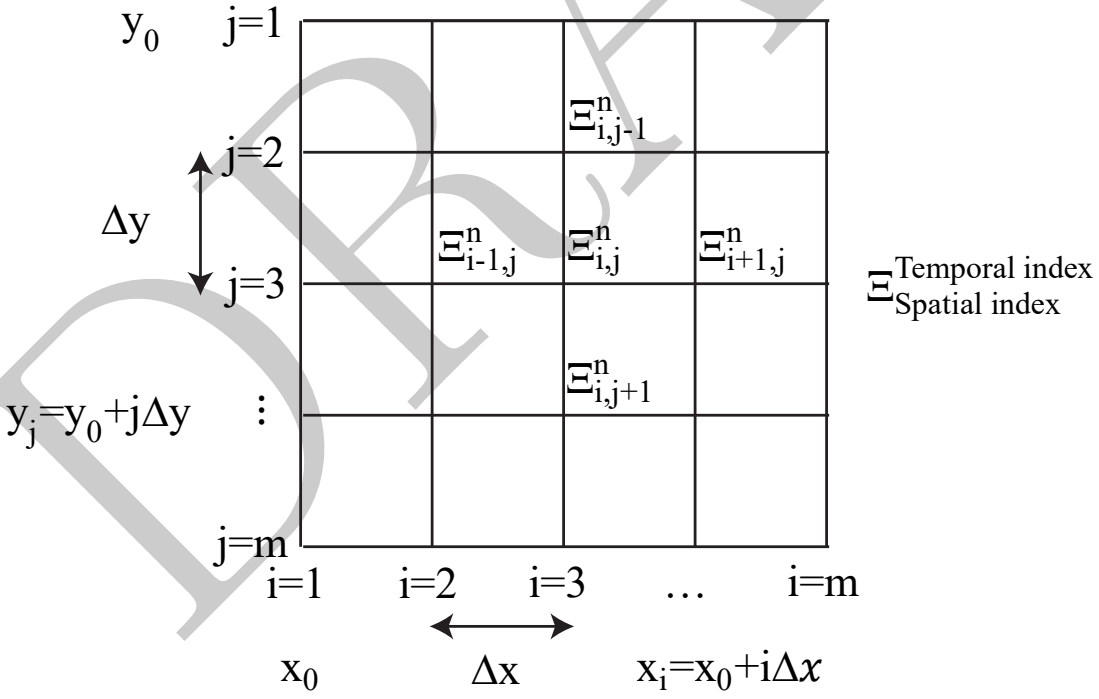


Figure 2: FDM discretized domain. Please be mindful of the indices. For example, notice that I am starting the j index (notwithstanding the domain defn in the previous figure) from the top left as in to keep semblance with the way MATLAB handles matrices.

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